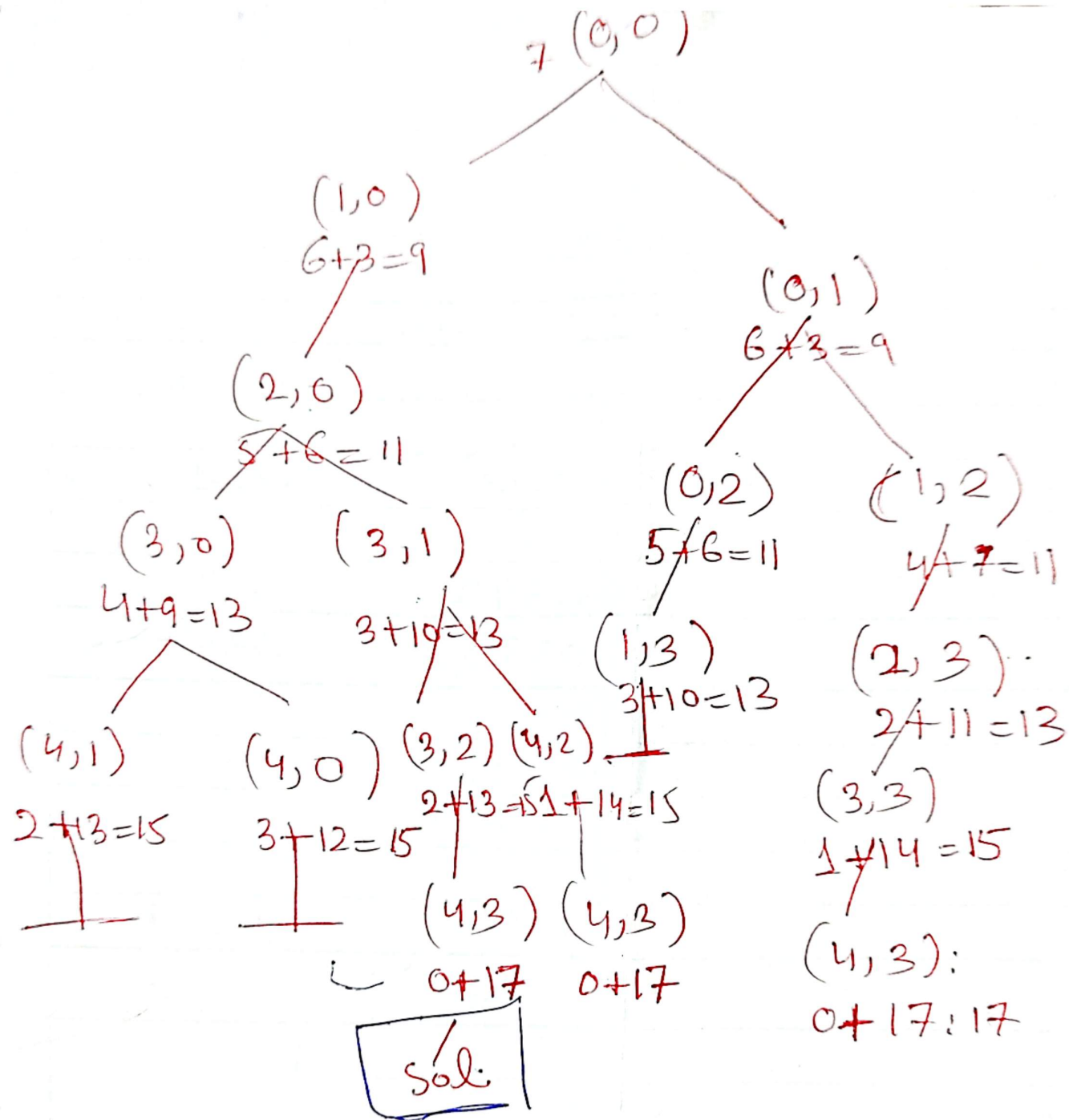




(0,0) → (1,0) → (2,0) → (3,1) → (3,2) → (4,3)

(0,0)

open

$$(0,1):9 \quad (1,0):9$$

$$\boxed{(0,0):9 \quad (0,2):11 \quad (1,2):11}$$

$$(0,2):11 \quad (1,2):11 \quad (2,0):11$$

$$(1,2):11 \quad (2,0):11 \quad (1,3):13$$

$$(2,0):11 \quad (1,3):13 \quad (2,3):13$$

$$(1,3):13 \quad (2,3):13 \quad (3,0):13 \quad (3,1):13$$

u //

$$(3,0):13 \quad (3,1):13 \quad (3,3):15$$

$$(3,1):13 \quad (3,3):15 \quad (4,1):15 \quad (4,0):15$$

$$(3,3):15 \quad (4,1):15 \quad (4,0):15 \quad (3,2):15$$

$$(4,2):15$$

$$(3,3):15 \quad (4,1):15 \quad (4,0):15 \quad (4,2):15$$

$$(4,3):17$$

$$(4,1):15 \quad (4,0):15 \quad (4,2):15$$

$$(4,3)^{33}:17 \quad (4,3)^{32}:17$$

$$(4,1):15 \quad (4,2):15 \quad (4,3)^{33}:17 \quad (4,3)^{32}:17$$

$$(4,2):15 \quad (4,3)^{33}:17 \quad (4,3)^{32}:17$$

$$(4,3)^{33}:17 \quad (4,3)^{(3,2)}:17 \quad (4,3)^{(4,2)}:17$$

$$(\text{goal}):17$$

close

$$\boxed{(0,0)}$$

$$\boxed{(0,0), (0,1)}$$

$$(0,0) \quad (0,1), (1,0)$$

$$(0,0) \quad (0,1) \quad (1,0) \quad (0,2)$$

$$(0,0) \quad (0,1) \quad (1,0) \quad (0,2) \quad (1,2)$$

$$+ (2,0)$$

$$+ (1,3)$$

$$+ (2,3)$$

$$(0,0) \quad (0,1) \quad (1,0) \quad (0,2) \quad (1,2) \quad (2,0)$$

$$(1,3) \quad (2,3) \quad (3,0)$$

$$+ (3,1)$$

$$+ (3,2)$$

$$+ (3,3)$$

$$(0,0) \quad (0,1) \quad (1,0) \quad (0,2) \quad (1,2) \quad (1,3) \quad (2,3)$$

$$(3,0) \quad (3,1) \quad (3,2) \quad (3,3) \quad (4,0)$$

$$+ (4,1)$$

$$+ (4,2)$$

$$(4,3)^{(3,2)}$$

$$\begin{array}{lll}
 h(0,0) = 7 & h(4,0) = 3 & h(4,2) = 1 \\
 h(1,0) = 6 & h(1,2) = 4 & h(4,1) = 2 \\
 h(0,1) = 6 & h(1,3) = 3 & h(3,1) = 3 \\
 h(2,0) = 5 & h(2,3) = 2 & h(4,3) = 0 \\
 h(0,2) = 5 & h(3,3) = 1 & \\
 h(3,0) = 4 & h(3,2) = 2 &
 \end{array}$$

Part e: $h(0,0) = 4 \times 3 + 3 \times 1 = \cancel{15} \Rightarrow 17$

$$h(1,0) = 4 \times 3 + 3(0) = 12 > (1,0) > (4,3)$$

Tree: 4 Path cost: 1.5
 $g(n)$ $h(n)$: 1.5 $f(n) = 2$

b: 2 open, close

c: 1 path Sol

d: 2 h_1

e: 2 h_2

f: 3 both h_1 & h_2